Teknik komputasi

Soal: untuk dua buah persamaan $f(x) = x^2 - x - 2 = 0$ dan $g(x) = x^2 - 4 = 0$,

Hitunglah pendekatan untuk nilai x sampai iterasi ketiga metode regulasi

: $f(x) = x^2 - x - 2$

$x_l = 1$

$x_u = 3$

$$x_r = x_u - \frac{f(x_u)(x_u - x_l)}{f(x_u) - f(x_l)}$$

$f(1) = -2$

$f(3) = 4$

iterasi 1

$$x_r = 1 - \frac{(-2)(3-1)}{4-(-2)} = 1 + \frac{4}{6} = 1,6667$$

$f(1,6667) = (1,6667)^2 - 1,6667 - 2$

$= 2,7778 - 1,6667 - 2$

$= 1,111 - 2 = -0,8889$

iterasi 2

$$x_r = 1,6667 - \frac{(-0,8889)(3-1,6667)}{4-(-0,8889)} = 1,9091$$

$f(1,9091) = (1,9091)^2 - 1,9091 - 2$

$= \cancel{1,69}0$

$= 3,6446 - 1,9091 - 2$

$= 1,7355 - 2 = -0,2645$

iterasi 3

$$x_r = 1,9091 - \frac{(-0,26)(3-1,9091)}{4-(-0,26)} = 1,9757$$

$f(1,9757) = (1,9757)^2 - 1,9757 - 2$

$= -0,0723$

$x_l = x_r$

iterasi	xl	xu	xr	f(xr)
1	0	3	1,333	
2	1,333	3	1,8462	
3	1,8462	3	1,9683	

$x_l = x_r$

iterasi	xl	xu	xr	f(xr)
1	1	3	1,6667	-0,8889
2	1,6667	3	1,9091	-0,26
3	1,9091	3	1,9757	-0,07

$x = 2$ (karena akar persamaan mendekati 2)

$g(x)$

$x_l = 0$

$x_u = 3$

$g(0) = -4$

$g(3) = 5$

iterasi 1

$x_r = 1,3333$

$$x_1 = 0 - \frac{(-4)(3-0)}{5-(-4)}$$

$$= 0 - \frac{(-4)(3)}{9}$$

$$= 0 - \frac{-12}{9} = 0 + 1,333$$

$$= 1,333$$

$g(x)$

$g(1,333) = (1,3333)^2 - 4$

$= 1,7778 - 4$

$= -2,2222$

$x_l = x_r = 1,3333$

iterasi 2

$$x_r = 1,3333 - \frac{(-2,2222)(3-1,3333)}{5-(-2,2222)}$$

$= 1,8462$

$g(x)$

$g(1,8462) = (1,8462)^2 - 4$

$= -0,5915$

$x_l = x_1 = 1,8462$

iterasi 3

$$x_1 = 1,8462 - \frac{(0,5915)(3-1,8462)}{3-(-0,5915)}$$

$= 1,9682$

$g(x)$

$g(1,9682) = (1,9682)^2 - 4$

$= -0,1262$

iterasi	x_l	x_u	x_r	$g(x_r)$
1	0	3	1,333	-2.2222
2	1,3333	3	1,8462	-0,5915
3	1,8462	3	1,9682	-0.1262

x_r mendekati 2